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KWAME NKRUMAH UNIVERSITY OF SCIENCE AND TECHNOLOGY, KUMASI, GHANA

AE 351

Soil and Water Engineering I

Lecture 6

ING PROF. N. KYEI-BAFFOUR, PhD

DEPARTMENT OF AGRICULTURAL ENGINEERING,
COLLEGE OF ENGINEERING, KNUST,
KUMASI-GHANA

nkyeibaffour@hotmail.com //+233 (0)24 4580 479

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BASIC SOIL MECHANICS: CONCEPTS AND APPLICATIONS IN ENGINEERING AND AGRICULTURE

- Soil mechanics is the application of laws of mechanics and hydraulics to engineering problems dealing with sediments and other unconsolidated accumulations of solid particles produced by the mechanical and chemical disintegration of rocks regardless of whether or not they contain an admixture of organic constituent.
- The objective of studying soil mechanics in agricultural engineering is to present the basic scientific principles influencing the nature and behaviour of soils as well as to discuss those management aspects which are particularly relevant to engineering in areas such as soil and water engineering.

Plants have roots and the roots are anchored in the soil. Development of plants depends on three soil factors:

- **Soil Type**: Texture (the relative sizes of soil mineral components)
- **Soil Structure**: Shows how soils are bonded together
- **Soil Fertility**: This largely depends on soil organic matter.

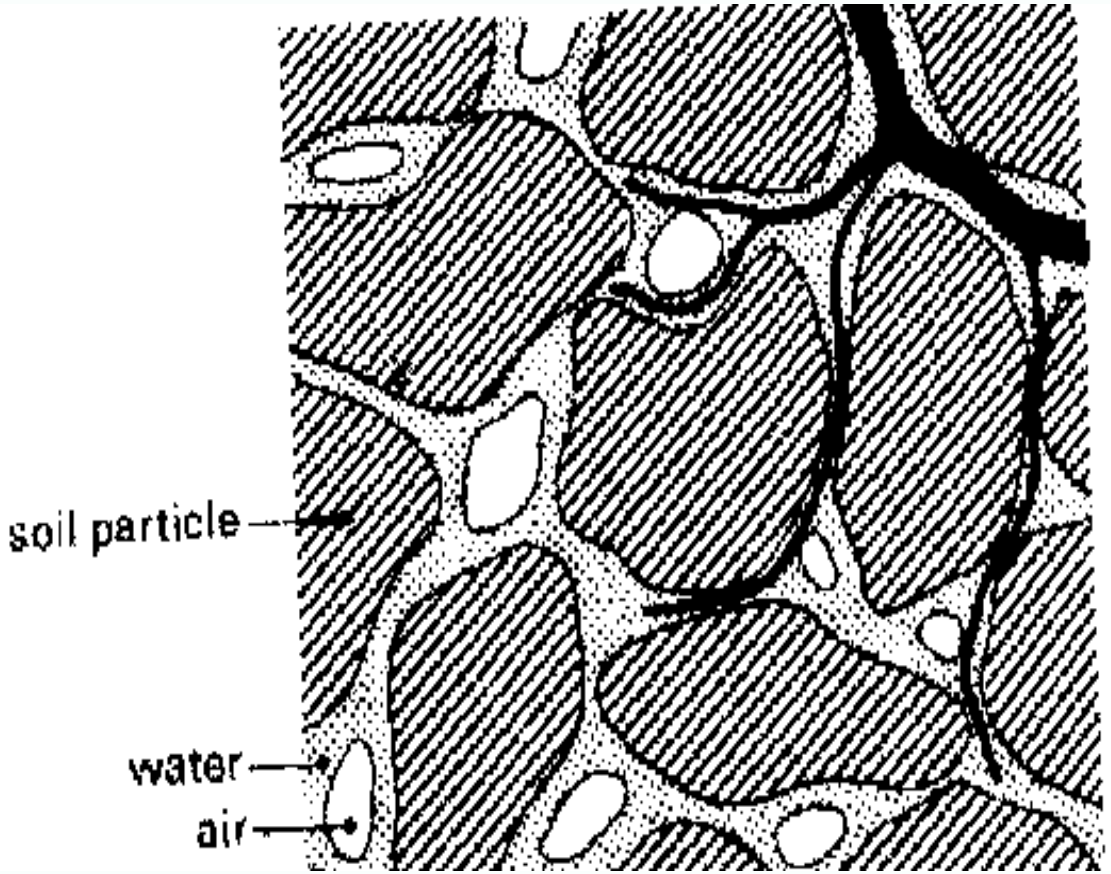
There are four basic soil constituents:

- (a) Mineral Material: Sand, clay and silt
- (b) Organic matter – (very valuable as nutrient source and binding soil particles together)
- (c) Water (Minerals are dissolved in this most soluble liquid)
- (d) Air (Needed for respiration of plant roots)

Soil Texture and Structure

- Soil texture and soil structure are very important because they determine the kind of crops to be grown, whether there will be erosion, whether to apply water to crops, the kind of machinery to use and so on.
- Organic matter from plants and animals helps to build up or improve soil structure.
- Organic matter also helps to improve the water holding capacity of the soil.
- The soil type can be controlled using farm machinery.
- Soil structure can be changed with soil machinery
- Soil structure studies are important in order to determine the amount of load and the type of machinery to be used in a particular soil.

Soil Constituents



Soil Constituents Continued

- The mineral matter and organic matter make up the solid contents of the soil while the water and the air represent the soil pores.
- From a structural point of view, air is irrelevant but it helps in aeration for plant growth.
- In an average agricultural soil, half will be solid solids, 45% minerals and about 5% organic matter. The other 50% which is called soil pores or voids will occupy 50%.
- Voids are dynamic and there can be 20–30% air and 30–20% water by volume.
- Excess water will lead to soil saturation and will cause death of most plants as poor aeration is obtained.

Mineral (Inorganic) Components of the Soil

- Except in the case of organic soils, most of a soil's solid framework consists of mineral particles. Quartz is the most common constituent of sand and silt.
- They are variable in size and composition. They can vary from small rock particles to colloids.
- The mineral can be raw (quartz and other primary materials – coarse fractions which have not changed from parent material)

- They can also be silicate clays and iron oxides formed by the breakdown and weathering of less resistant minerals as soil formation progressed. These are called secondary minerals. The common clay minerals tend to form in sheet or plate like structures are mainly montmorillonife or smectite, illite, and kaolinite or kaolin.
- The mineral particles present in soils are extremely variable in size.
- The larger soil particles, which include stones, gravel and coarse sands, are generally rock fragments of various kinds.
- Excluding the larger rock fragments such as stones and gravel, soil particles range in size over four orders of magnitude: from 2 mm to less than 0.002 mm in diameter.

Soil Particles

Sand Particles

- Sand particles are visible to the naked eye and vary in size from 2.0-0.05 mm.
- They are gritty when rubbed between fingers. Sand particles do not adhere to one another and are therefore not sticky.

Silt and Clay Particles

- Silt particles are smaller than sand and range from 0.05-0.002 mm.
- The silt particles are too small to be seen without a microscope. It feels smooth but not sticky, even when wet.
- Clays are the smallest class of mineral particles being < 0.002 mm.
- They adhere together to form a sticky mass when wet and form hard clods when dry.

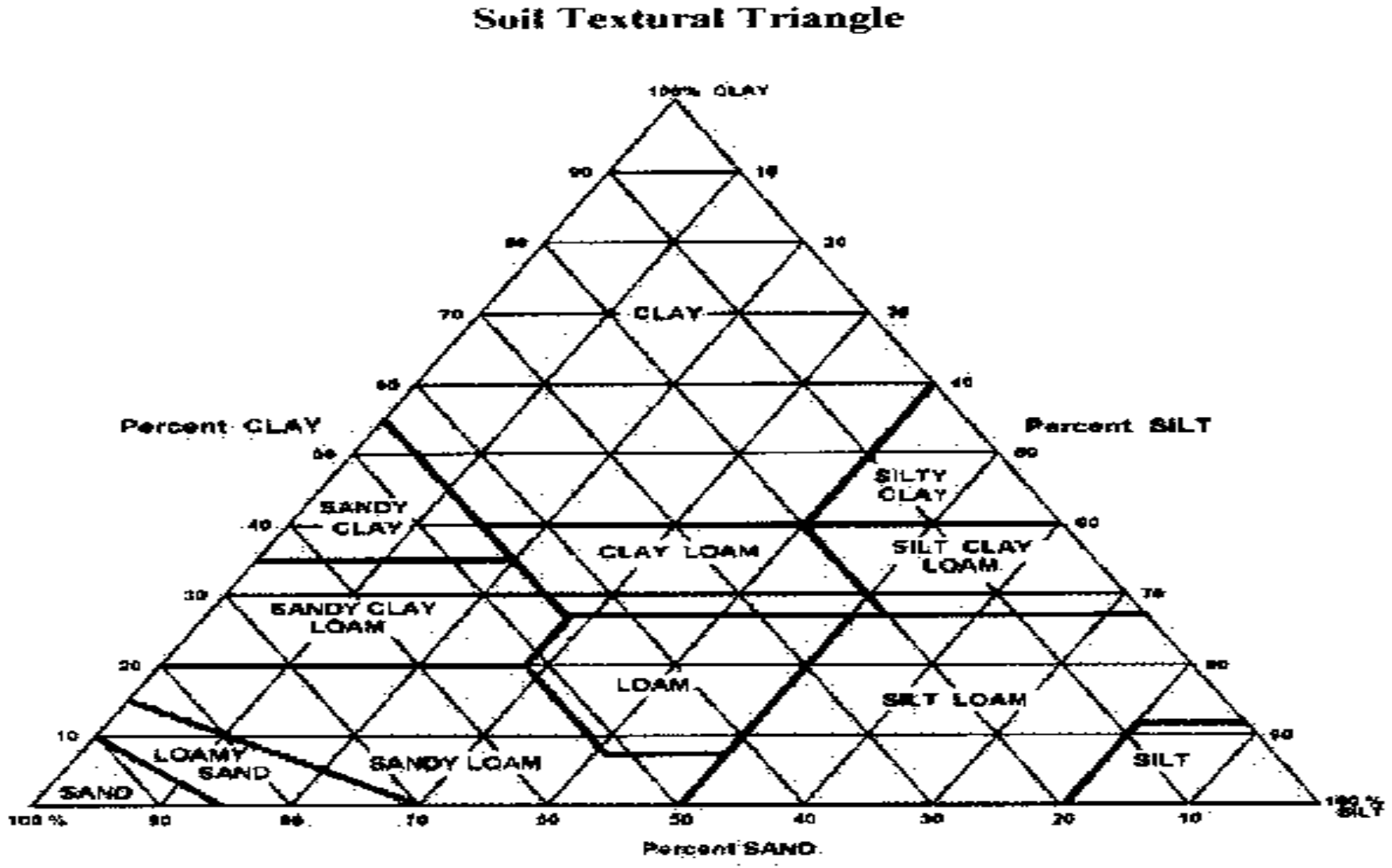
Classification of Soil Constituents

Primary Particle	USDA (United States Department of Agriculture)	ISSS (International Soil Science Society)
ROCKS	> 2 mm	> 2 mm
SAND	0.05 to 2 mm	0.02 to 2 mm
SILT	0.002 to 0.05 mm	0.002 to 0.02 mm
CLAY	< 0.002 mm	< 0.002 mm

Soil Texture

- This refers to the relative proportions of the various soil separates (sand, silt and clay) in a soil. Terms such as sandy loam, silty clay, and clay loam are used to identify soil texture using the USDA textural triangle.
- There are two methods that are generally utilized to determine the particle size distribution of soil: These are Sieve Analysis (for particle sizes > 0.075mm in diameter) and hydrometer Analysis (< 0.075mm). Particle size analysis is performed by shaking the soil sample through a set of sieves having progressively smaller openings.
- Dry sieving is no longer used
- For silt and clay, separation is by the rate of settling (Stoke's Law) which can be obtained in a pipette or hydrometer analysis. They are put in a solution and the time of settling is noted. From the Mechanical analysis, the proportions of sand, silt and clay are obtained. The actual soil texture is determined using the Soil Textural Triangle e.g. For a soil with 50% sand, 20% silt and 30% clay, the texture is Sandy Clay Loam.

USDA Soil Textural Triangle



Colloidal Material

- The smaller particles (< 0.001 mm) of clay (and similar sized organic particles) have colloidal properties and can be seen with an electronic microscope.
- The colloidal particles have a very large area per unit weight so there are enough surface charges to which water and ions can be attracted.
- These charges make them adhere together. Humus improves the water holding capacity of the soil.

Water

- Water is held in the soil at various degrees of tenacity which varies with the amount of water present.
- The tension force of water in unsaturated soils has been described by several expressions such as soil-pull, the force of suction and capillary tension. Suction can be defined as the force per unit area that must be exerted to remove water from the soil. Suction or tension is measured in bars or Pascals.
- When the soil is saturated or nearly so, the amount of suction is almost zero, but as the soil water depletes, greater amount of energy must be applied to extract water.
- The soil can be initially saturated and if it is drained, field capacity can be reached. Some water is still held by surface tension.

Water in the Soil

Field Capacity

- Field Capacity is the greatest amount of water the soil can hold under drainage. For most soils, it is obtained after two days of drainage after the soil was saturated by heavy rain or irrigation. It is the optimum amount of water needed for agriculture.

- **Permanent Wilting Point and Available Water**
- Below field capacity, the plant finds it more and more difficult to extract water until the suction or tension reaches 15 bars **permanent wilting point is obtained**, which is the maximum tension the plant can exert on the soil to extract water.
- Available water is the difference between the moisture contents at field capacity and permanent wilting point.
- Clay holds more water but the plants exert more tension to extract water more than sand.

Aeration

Soil Air

- From a mechanical point of view, air is irrelevant as it cannot support load. Soil air is different from atmospheric air. It is found in voids around the soil solids.
- Soil air has a very high relative humidity when compared with atmospheric air
- It has high carbon dioxide levels and low oxygen levels in comparison with atmospheric air

Aeration Effects

Aeration affects root development by three main factors

- By oxygen content: There is restricted root development if oxygen falls below 9%. If it drops to less than 5%, root development ceases.
- Carbon dioxide content: the roots will survive up to 9 to 10%. The optimum is < 1%.
- By-products of anaerobic decomposition: Hydrogen sulphide is very toxic at low concentrations and kills roots. Methane and hydrogen are all right at reasonable concentrations.

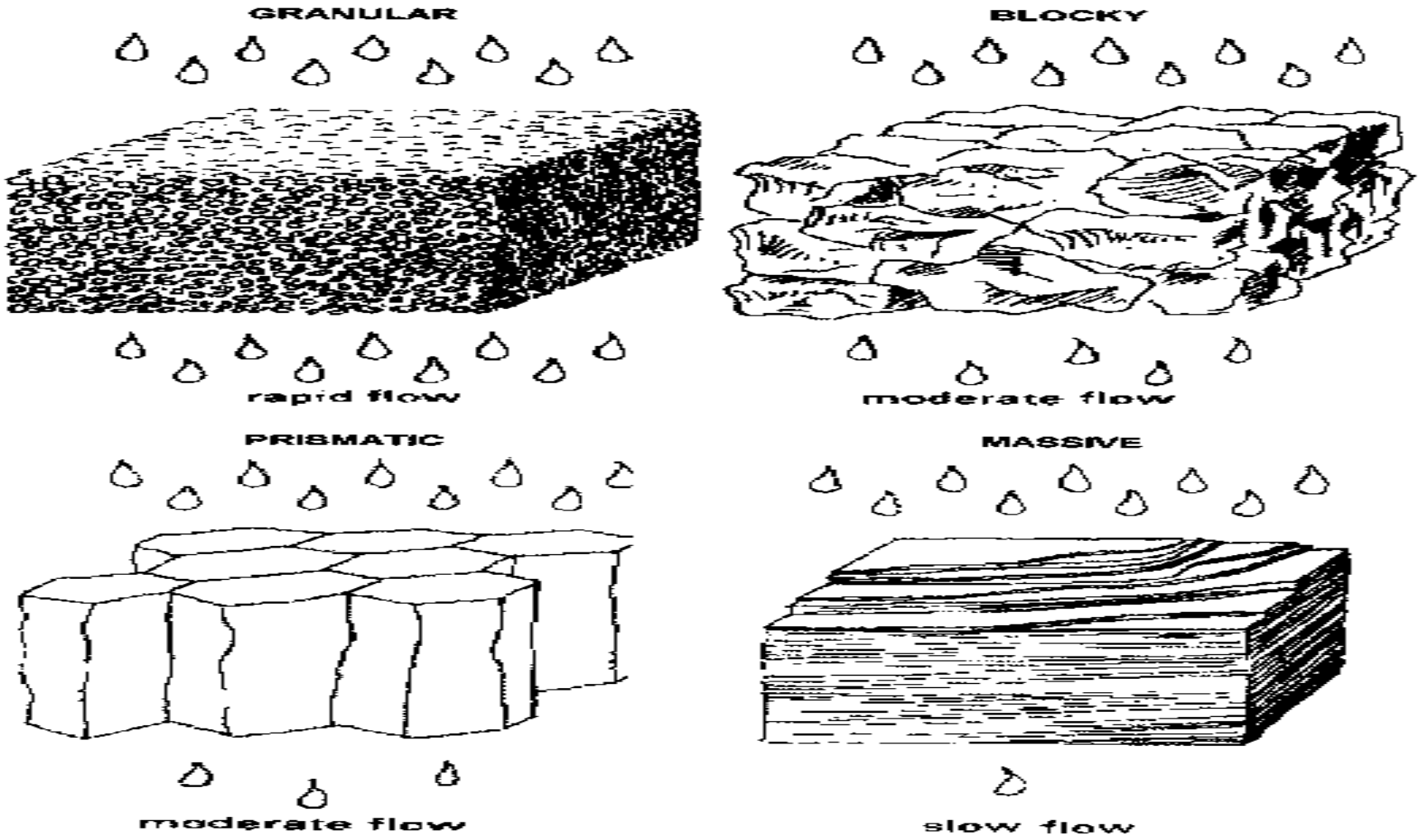
SOIL STRUCTURE

- This is a field term that describes the aggregation of the soil primary particles.
- Managing soil is managing soil structure.

The soil structural types include:

- **Massive Structure:** No aggregation e.g. big block of clay or mass of sand grains. In clay, the more the dryness, the deeper the cracks from the surface and the following structure is obtained:
- **Prismatic or Columnar Structure :** Caused by vertical cracking
- If there is secondary horizontal cracks, the following structure is obtained:
- **Blocky Structure:**
- If the blocky structure is worked upon by man, the spheroidal or granular or crumb structure is obtained.
- **Plate-like Structure**

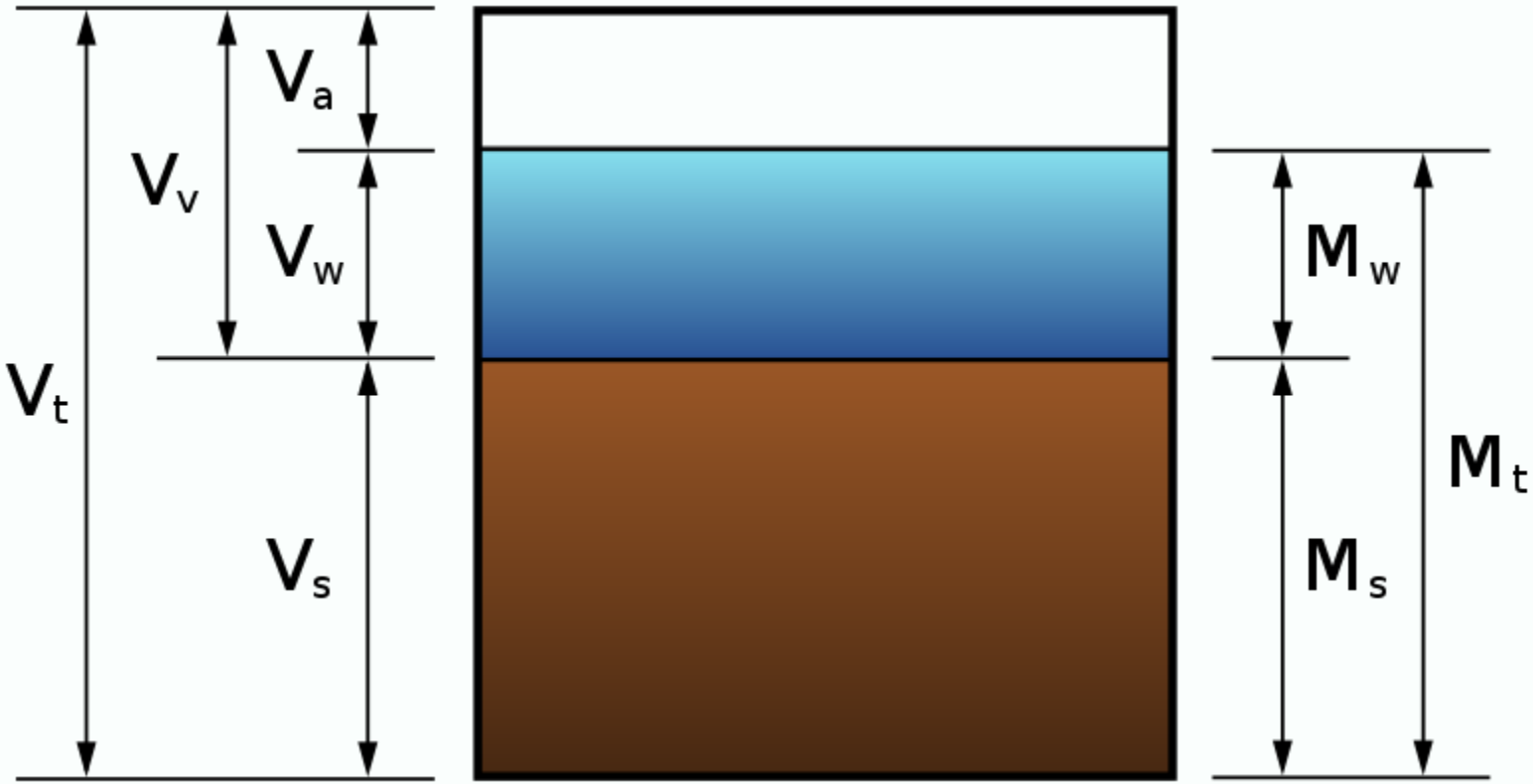
Flow of Water through Different Soil Structures



REPRESENTATIVE PHYSICAL PROPERTIES OF SOILS

Soil Texture	*Infiltration [a] and Permeability (cm/h)* I_f	*Total Pore Space (%)* n	*Apparent Specific Gravity (Dry Bulk Density)* ρ_b	*Field Capacity (%)* θ_{FC}	*Permanent Wilting Point (%)* θ_{PWP}	*Total Available Moisture**		
						Dry Weight (%) $\theta_w = \theta_{FC} - \theta_{PWP}$	*Volume (%)* $\theta_v = \theta_w \rho_b$	*cm/m* $D = \rho_b D\theta_w/100$
	5 (2.5-25)	38 (32-42)	1.65 (1.55-1.80)	6 (6-12)	4 (2-6)	5 (4-6)	8 (6-10)	8 (7-10)
Sandy Loam	2.5 (1.3-7.6)	43 (40-47)	1.50 (1.40-1.60)	14 (10-18)	6 (4-8)	8 (6-10)	12 (9-15)	12 (9-15)
Loam	1.3 (0.8-2.0)	47 (43-49)	1.40 (1.35-1.50)	22 (18-26)	10 (8-12)	12 (10-14)	17 (14-20)	17 (14-19)
Clay Loam	0.8 (0.25-1.5)	49 (47-51)	1.35 (1.30-1.40)	27 (23-31)	13 (11-15)	14 (12-16)	19 (20-22)	19 (17-22)
Silty Clay	0.25 (0.03-0.5)	51 (49-53)	1.30 (1.30-1.40)	31 (27-35)	15 (13-17)	16 (14-18)	21 (18-23)	21 (18-23)
Clay	0.5 (0.01-0.1)	53 (51-55)	1.25 (1.20-1.30)	35 (31-39)	17 (15-19)	18 (16-20)	23 (20-25)	23 (20-25)

SOIL PHASE RELATIONS



Mass-Volume Relationship of Soils

- V_a is the Volume of air,
- V_w is the Volume of water
- V_s is the Volume of solids, V_v is the Volume of voids or pores (water & air)
- V_t is the total volume (air, water and solids)
- M_a is the Mass of air which is approximately zero,
- M_w is the Mass of water
- M_s is the Mass of solids; M_v is the Mass of voids or pores (water & air)
- M_t is the total mass (air, water and solids)

Definitions

- **Density of Solid Particles, ρ_s (Particle density):**
- **(a)** It is defined as the mass per unit volume of soil solids usually expressed as g/cm^3 or Mg/m^3.
- i.e. Particle density = M_s/V_s where M_s is the mass of dry soil
- Particle density is essentially the same as the specific gravity of a soil substance and varies within a narrow range of 2.60-2.75 g/cm^3 with the average being assumed as 2.65 g/cm^3 for a typical soil with 1-5 % organic matter content.
- Montmorillonite can have a particle density of 2.74 g/cm^3, quartz, 2.65 and kaolinite, 2.61 g/cm^3.

Dry Bulk Density, ρ_b

- It is defined as the mass of a unit volume of dry soil. This includes both solids and pores. i.e. Bulk density = M_s/V_t ; M_s is the mass of dry soil and V_t is the total volume of undisturbed soil ***in situ***.
- The major method of measuring bulk density in the field is to collect a known volume of undisturbed soil (V_t) in a soil core, and drying it in the oven at a temperature of 105 °C for 24 h to remove all the water to obtain M_s. The values of dry bulk density range from 1.0 for loose open soil to about 1.8 g/cm^3 for compacted soil.
- Values of bulk density are mainly affected by soil texture (sandy soils have more density than silty and clay soils), degree of soil aggregation and is reduced by soil organic matter content.

Porosity

- It is defined as the proportion of the volume of soil pores (air and water) in comparison with the total volume of soil.
- i.e. Porosity, n = Volume of Pores, V_v/V_t
- It normally ranges from 0.2 (20%) to 0.6 (60%).
- One of the main reasons for measuring soil dry bulk density is that this value can be used to calculate soil porosity.
- Give the same particle density, the lower the dry bulk density the higher the porosity.

Soil Pores

- **Void Ratio (e):** It is defined as the volume of Pores (V_v) divided by the volume of solids, V_s. It can be shown that:

- **Degree of Saturation, S:** This is the volume of water divided by the volume of voids. A soil is said to be saturated when all pores are filled with water i.e. when:
- $V_w = V_v$

- **Soil Wetness**

This can be expressed by mass or by volume.

- **By Mass (θ_m)** – gravimetric system: Weigh soil, put in oven at 105°C for about 24 hours; weigh again and obtain the mass of water by subtraction.
- A good soil should have moisture contents between 5 and 60% and for peat or organic soils; it can be greater than 100%.
- **Volumetric water content, $\theta_v = \theta_m \rho_b$**

First Sample Question with Solution

- The volume fraction of water in a field is 0.12. Find the quantity of water (m^3) required to raise 0.8 m of soil depth to a volumetric field capacity (FC) water content of 0.30 over an irrigated area of 64 ha?

Solution

Depth of water = (0.30-0.12) x 0.8 m = 0.144

Area of land =64 x 104 m^2

Quantity of water = 0.144 m x 64 x 104 m^2

= 92,160 m^3

Second Sample Question with Solution

A cube of moist soil of side 10 cm is taken from the field. The total mass of the sample (M_t) is 1.50 kg of which 0.25 kg is water (M_w). The soil particle density (ρ_s) is 2650 kg/m^3 and the density of water (ρ_w) is 1000 kg/m^3. Find the following:

a) Mass fraction of water(θ_m)?

b) Volume fraction of water (θ_v)?

c) Volume percentage of water?

d) Depth of water in mm/m soil depth (D)?

e) Dry bulk density (ρ_b)?

f) Soil porosity (n)?

g) Air-filled porosity (n_a)?

Solution

a) Mass fraction of water (θ_m) $= M_w/M_s = 0.25/(1.50-0.25)$

$= 0.20$ or 20 %

b) Volume fraction of water (θ_v) $= M_w/\rho_w/$Volume of cube

$= 0.25/1000/0.1 \times 0.1 \times 0.1$

$= 0.25$ or 25 %

c) Volume percentage of water (θ_v) $= 0.25 \times 100$ % = 25 %

d) Depth of water in mm/m soil depth (D)

$= 25$ % $= 25/100$

$= 25/100/(250/1000)$ (mm/mm)

Solution continued

e) Dry bulk density, $\rho_b = M_s/V_t$

$= (1.50\text{-}0.25)\ kg/(0.1 \times 0.1 \times 0.1)\ m^3$

$= 1250\ kg/m^3$

Soil bulk density relative to water $= \rho_b/\rho_w$

$= 1250/1000 = 1.25$

f) Soil porosity (n) $= 1 - \rho_b/\rho_s = 1 - 1250/2650$

$= 0.53$ or 53 %

Solution continued

g) Air-filled porosity (n_a) = Soil porosity – θ_v
= 0.53-0.25 = 0.28

Alternatively,

Air-filled porosity = Air-filled pore volume/Bulk volume of soil

Volume of solids, V_s = M_s / ρ_s = 1.25/2650 = 0.00047 m^3

Volume fraction of solids = Volume of solids/Bulk volume of soil
= 0.00047 m^3/0.001 m^3 = 0.47

Therefore,

Air-filled porosity (n_a) = 1- Volume fraction of solids - θ_v
= 1-0.47-0.25 = 0.28

Third Sample Question with Solution

A soil of volume 290 cm^3 and mass 190 g is used to take an undisturbed soil sample in a proposed vegetable garden at KNUST. The mass of the sample and core was 730 g. The mass of the same sample and core at field capacity was 790 g. The sample was then dried in an oven at 105 °C for 24 h and then weighed again. The mass of the dry soil sample and core was 680 g. Given that the PWP of the soil is 8 % and the soil allowable moisture depletion is 50 %, determine the following:

a) Dry bulk density (ρ_b)?

b) Field moisture content on weight basis (θ_w)?

c) FC on weight basis (θ_{FC})?

d) Available soil moisture per metre depth of soil?

e) Using an effective design rooting depth (D) of 1 m, compute the net water requirement (NWR) for the vegetables?

f) Assuming an irrigation application efficiency (η) of 60 % for the envisaged irrigation method, calculate the gross water requirement (GWR)?

g) If the peak and minimum consumptive use of the vegetables in the area are 12.5 and 4.5 mm/d, what will be the irrigation interval?

Solution

Let core Volume, V_b	= 290 cm^3
Core mass, M_1	= 190 g
Mass of field sample and core, M_2	= 730 g
Mass of field sample at FC, M_3	= 790 g
Mass of dry sample and core, M_4	= 680 g
PWP	= 8 %
MAD	= 50 %

Solution Continued

a) $\rho_b = M_4 - M_1/V_b = (680\text{-}190)\ g/290\ cm^3$

$= 1.69\ g/cm^3$

Soil bulk density relative to water $= \rho_b/\rho_w$

$= 1.69/1.00 = 1.69$

b) Field moisture content $= (M_2\text{-}M_4)\times 100\ \%/(M_4\text{-}M_-)$

$= (730\text{-}680)\times 100\ \%/(680\text{-}190)$

$= 10.2\ \%$

c) FC moisture content $= (M_3\text{-}M_4)\times 100\ \%/(M_4\text{-}M_-)$

$= (790\text{-}680)\times 100\ \%/(680\text{-}190)$

$= 22.45\ \%$

Solution Continued

d) Available moisture content, FC – PWP = 22.45-10.2 = 14.45 %

Available soil moisture (volumetric) = 14.45 % x 1.69 = 24.42 %

= 24.42/100 x10 mm/10 mm = 244.2 mm/m depth of soil

e) NWR = 10 x Available moisture x ρ_b x D(m) x MAD

= 10 x (FC-PWP) x ρ_b x D(m) x MAD

= 10 x (22.45-8) x 1.69 x 1 m x 0.5

= 122 mm

Solution Continued

f) GWR = NWR/ n_a = 122 mm/60 % = 122/0.6 = 203 mm

g) Irrigation interval = NWR (mm)/ Peak consumptive use (mm/d)
= 122 mm/12.5 mm/d = 10 d

Thank You

For any concerns, please contact

elearning@knust.edu.gh
elearningknust@gmail.com
0322 191132

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